

Title: Efficacy of Underwater EMR for non-pedunculated colorectal lesions: a systematic review and meta-analysis protocol

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Author contributions:

TY is the guarantor. TY, NH, TS, MB, JW and HM drafted the manuscript.

All authors contributed to the development of the selection criteria, the risk of bias assessment strategy and data extraction criteria. TY, NH, TS, MB and JW developed the search strategy. TY, NH, TS, MB and JW provided statistical expertise. MB and JW provided expertise on TY. All authors read, provided feedback and approved the final manuscript.

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1.Introduction

Colorectal cancer is the third most common cancer in the world and the fourth most common cause of cancer-related mortality ¹⁻³. However, if detected and treated at an early stage, a favorable prognosis is expected. Endoscopic mucosal resection (EMR) is widely used in the prevention of colorectal cancer. It is performed by injecting normal saline into the submucosal layer between the lesion and the muscle layer to protrude the mucosa and then entrapped with a snare and resection using electrocautery ⁴. However, some proficiency in the method of injection and snaring is required, and The en bloc resection rate decreases when the polyp size exceeds 10 mm. The resulting high local recurrence rate after piecemeal resection is clinically problematic ^{5,6}.

Recently, underwater EMR (UEMR) has emerged by Bimmoeller et al.⁷ as an alternative to conventional EMR (CEMR) and is reported to be effective for removing flat or large colorectal polyps. With UEMR, the bowel lumen is filled with water instead of air, and the lesion is captured and resected with a snare without submucosal injection of normal saline. Compared to CEMR, it is easier to snare large colorectal polyps, residual recurrent lesions, and depressed lesions, and has the potential to improve treatment outcomes. RCTs compared to CEMR have shown significantly higher R0 resection rates for UEMR ⁸ and significantly shorter procedure time than CEMR ⁹.

Although a meta-analysis of UEMR has shown efficacy and safety, the superiority of UEMR over CEMR has not been sufficiently examined because it was not compared with CEMR ¹⁰ or because it was compared with CEMR in a mixture of cohort studies and RCTs ¹¹. Therefore, we exclude cohort studies and conduct a meta-analysis with additional RCTs.

2.Research question

The effectiveness of UEMR compared with CEMR for endoscopic removal of sessile colorectal lesions.

P: Colorectal lesion

I: UEMR

C: CEMR

O: R0 resection

3.Method

3.1 Protocol

We used a systematic review protocol template ¹². We followed the Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 for preparing this protocol ¹³. We will publish this protocol in protocols.io (<https://www.protocols.io/>).

3.2 Inclusion criteria of the articles for the review

3.2.1 Type of studies

We will include randomized controlled trials that assess the effectiveness of UEMR compared with CEMR for endoscopic removal of colorectal lesions. We will not apply language or country restrictions. We will include all papers including published, unpublished articles, abstract of conference and letter.

We will exclude duplicate publication, cluster randomized trial, cross-over study, quasi-experimental design, case report and non-human studies. We will not exclude studies based on the observation period or publication year.

3.2.2 Study participants

All patients who underwent UEMR or CEMR for colorectal lesions.

Inclusion criteria:

Adult patients (> 18 years of age) with colorectal lesion

Exclusion criteria:

- 1) Residual and recurrent lesions after endoscopic resection
- 2) Patients with colitis ulcer, Crohn disease, familial adenomatous polyposis.

3.2.3 Intervention

Underwater EMR: complete deflation of the colorectal lumen air and total immersion of the lesion in normal saline. Then snaring the lesion and the surrounding mucosa and resection using electrocautery.

3.2.4 Control

CEMR: needle injection of normal saline into the submucosa and then entrapment of the mucosal protrusion with a snare and resection using electrocautery.

3.3 Type of outcomes

3.3.1 Primary outcomes

1. Proportion of R0 resection

Definition: proportion of R0 resection is calculated as the number of R0 resection divided by the total number of lesions. R0 resection was defined as en bloc resection with a histologically confirmed negative resection margin. En bloc resection was defined as endoscopically assessed removal of the lesion in one piece.

Period:

2. Mean procedure time

Definition: The procedure time in the UEMR group was measured from the start of immersion in normal saline from the endoscope until complete removal of the polyp. The procedure time in the CEMR group was measured from insertion of the injection needle until complete removal of the polyp.

Period:

3.3.2 Secondary outcomes

1. Proportion of en block resection

Definition: proportion of en block resection is calculated as the number of en block resection divided by the total number of lesions. En bloc resection was defined as endoscopically assessed removal of the lesion in one piece.

Period:

2. All adverse events

Definition: Definition of adverse events is set by original authors.

Period: during follow up period

3.4 Search method

3.4.1 Electronic search

We will search the following databases:

1. the Cochrane Central Register of Controlled Trials (CENTRAL);
 2. MEDLINE via PubMed;
 3. EMBASE via ProQuest Dialog;
- See Appendix 1, 2, and 3 for the search strategies.

3.4.2 Other resources

We will also search the following databases for ongoing or unpublished trials:

1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP);
2. ClinicalTrials.gov;

See Appendix 4, 5 for the search strategies.

We will check the reference lists of studies, including international guidelines (published by the Japanese Society for Cancer of the Colon and Rectum (JSCCR), the European Society of Gastrointestinal Endoscopy and the U.S. Multi-society Task Force on Colorectal Cancer)¹⁴⁻¹⁶ as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of the studies

Two independent reviewers (TY and NH) will screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. We will contact original authors if relevant data is missing. Disagreements between the two reviewers will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (TS).

3.5.2 Data extraction and management

Two reviewers (TY and NH) will perform independent data extraction of the included studies using standardized data collection form. We will use a pre-checked form using 10 randomly selected studies.

The form will include the information on study design, study population, patient characteristics (sex, mean age and number of patients with UEMR and CEMR, number of lesions, mean lesion size, lesion location, lesion histological type), R0 resection rate with UEMR and CEMR, en bloc resection rate with UEMR and CEMR, mean procedure time with UEMR and CEMR, and adverse event rate (bleeding and perforation). Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (TS).

3.6 Assessment of risk of bias in included studies

Two reviewers (TY and NH) will evaluate the risk of bias independently using the Risk of Bias 2¹⁷. Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer (TS) will be acting as an arbiter, if necessary.

3.7 Measures of treatment effects

We will pool the relative risk ratios and the 95% confidence intervals (CIs) for the following binary variables: R0 resection, en block resection.

We will pool the mean differences and the 95% CIs for the following continuous variables: procedure time of UEMR and CEMR

If several different scales have been used in the included studies, we will pool the effect estimates using standard mean differences (SMDs)

3.8 Unit of analysis issues

For continuous data, only the sample size will be reduced; means and standard deviation will remain unchanged.¹⁷

Randomised cross-over studies

We will consider only data from the first period.

Multiple comparisons

All intervention groups that are relevant to this review will be included.

3.9 Handling of missing data

We will ask not-presented data to the original authors.

3.9.1 Missing outcomes

We will perform the intention-to-treat (ITT) analysis for all dichotomous data as much as possible. For continuous data, we will not impute missing data based on the recommendation by Cochrane handbook¹⁸. We will perform meta-analysis about the available data in the original study.

3.9.2 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman¹⁹. If we don't know these values when we contact the authors, standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook¹⁸, or validated method¹⁹. Validity of these methods will be analyzed by sensitivity analysis.

3.10 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I² statistic (I² values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). When there is substantial heterogeneity (I² > 50%), we will assess the reason for the heterogeneity. Cochrane Chi² test (Q-test) will be performed for I² statistic, and P value less than 0.10 will be defined as statistically significant.

3.11 Assessment of reporting bias

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. We will assess the potential publication bias by visual inspection of the funnel plot. Egger test will be performed as well. We will not conduct the test when we find less than 10 trials or trials which have similar sample size. We will assess the potential publication bias by visual inspection of the funnel plot

3.12 Meta-analysis

Meta-analysis will be performed using Review Manager software (RevMan 5.4). We will use a random-effects model.

3.13 Subgroup analysis

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. (For participants) (lesion size $\leq 20\text{mm}$ vs. $> 20\text{mm}$)

3.14 Sensitivity analysis

We will undertake the following sensitivity analyses for the primary outcomes to assess whether the results of the review are robust to the decisions made during the review process.

1. Exclusion of studies using imputed statistics.
2. Missing participants: verify the robustness of the results by seeking informative missingness odds ratios²⁰.
3. Only the participants who complete the study with complete data

4. Summary of findings table

Summary of findings table will be made for the following outcome based on the Cochrane handbook¹⁸.

We will include grading to evaluate the quality of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach for each Summary of findings table²¹.

1. Proportion of R0 resection
2. Mean procedure time
3. Proportion of en bloc resection
4. All adverse events

5. Conflict of Interest

The authors declare no conflicts of interests.

References

1. Colorectal cancer statistics | World Cancer Research Fund International. <http://www.wcrf.org/int/cancer-facts-figures/data-specific-cancers/colorectal-cancer-statistics>. Accessed January 10, 2015.
2. WHO | The global burden of disease: 2004 update.
3. Global Burden of Disease Cancer Collaboration GB of DC, Fitzmaurice C, Dicker D, et al. The Global Burden of Cancer 2013. *JAMA Oncol.* 2015;1(4):505-527. doi:10.1001/jamaoncol.2015.0735
4. Tanaka S, Saitoh Y, Matsuda T, et al. Evidence-based clinical practice guidelines for management of colorectal polyps. *J Gastroenterol.* 2015;50(3):252-260. doi:10.1007/s00535-014-1021-4
5. Hurlstone DP, Sanders DS, Cross SS, et al. Colonoscopic resection of lateral spreading tumours: a prospective analysis of endoscopic mucosal resection. *Gut.* 2004;53(9):1334-1339. doi:10.1136/gut.2003.036913
6. Hotta K, Fujii T, Saito Y, Matsuda T. Local recurrence after endoscopic resection of colorectal tumors. *Int J Colorectal Dis.* 2009;24(2):225-230. doi:10.1007/s00384-008-0596-8
7. Binmoeller KF, Weilert F, Shah J, Bhat Y, Kane S. "Underwater" EMR without submucosal injection for large sessile colorectal polyps (with video). *Gastrointest Endosc.* 2012;75(5):1086-1091. doi:10.1016/j.gie.2011.12.022
8. Yamashina T, Uedo N, Akasaka T, et al. Comparison of Underwater vs Conventional Endoscopic Mucosal Resection of Intermediate-size Colorectal Polyps. *Gastroenterology.* 2019;157(2):451-461. doi:10.1053/j.gastro.2019.04.005
9. Yen AW, Leung JW, Wilson MD, Leung FW. Underwater versus conventional endoscopic resection of nondiminutive nonpedunculated colorectal lesions: a prospective randomized controlled trial (with video). *Gastrointest Endosc.* 2020;91(3):643-654.e2. doi:10.1016/j.gie.2019.09.039
10. Spadaccini M, Fuccio L, Lamonaca L, et al. Underwater EMR for colorectal lesions: a systematic review with meta-analysis (with video). *Gastrointest Endosc.* 2019;89(6):1109-1116.e4. doi:10.1016/j.gie.2018.10.023
11. Li P, Ma B, Gong S, Zhang X, Li W. Underwater endoscopic mucosal resection for colorectal lesions: a meta-analysis. *Surg Endosc.* 2020. doi:10.1007/s00464-020-07745-8
12. Yuki Kataoka, Yasushi Tsujimoto, Masahiro Banno, Shunsuke Taito, Ryuhei So, Jun Watanabe, Akihiro Shiroshita 2020. protocol_template_for_intervention_review. protocols.io dx.doi.org/10.17504/protocols.io.biqrkdv6
13. Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ.* 2015;349(jan02 1):g7647-g7647. doi:10.1136/bmj.g7647
14. Hashiguchi Y, Muro K, Saito Y, et al. Japanese Society for Cancer of the Colon and Rectum (JSCCR) guidelines 2019 for the treatment of colorectal cancer. *Int J Clin Oncol.* June 2019. doi:10.1007/s10147-019-01485-z
15. Ferlitsch M, Moss A, Hassan C, et al. Colorectal polypectomy and endoscopic mucosal resection (EMR): European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline. *Endoscopy.* 2017;49(03):270-297. doi:10.1055/s-0043-102569
16. Kaltenbach T, Anderson JC, Burke CA, et al. Endoscopic Removal of Colorectal Lesions—Recommendations by the US Multi-Society Task Force on Colorectal Cancer. *Gastroenterology.* 2020;158(4):1095-1129. doi:10.1053/j.gastro.2019.12.018
17. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ.* August 2019;l4898. doi:10.1136/bmj.l4898
18. Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for Systematic Reviews of Interventions version 6.0 (updated July 2019). Cochrane, 2019.
19. Furukawa T a, Barbui C, Cipriani A, Brambilla P, Watanabe N. Imputing missing standard deviations in meta-analyses can provide accurate results. *J Clin Epidemiol.* 2006;59(1):7-10.

- doi:10.1016/j.jclinepi.2005.06.006
20. Higgins JP, White IR, Wood AM. Imputation methods for missing outcome data in meta-analysis of clinical trials. *Clin Trials J Soc Clin Trials*. 2008;5(3):225-239.
doi:10.1177/1740774508091600
21. Guyatt G, Oxman AD, Akl E a, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol*. 2011;64(4):383-394.
doi:10.1016/j.jclinepi.2010.04.026

Appendix 1: CENTRAL search strategy

- #1 MeSH descriptor: [polyps] explode all trees
#2 MeSH descriptor: [neoplasms] explode all trees
#3 MeSH descriptor: [intestines] explode all trees
#4 ((colorect* or colon* or bowel or rect* or intestinal mucosa) and (polyp* or neoplas* or tumour* or tumor or adenom* or lesion* or carcinom* adenocarcinom* or cancer*)):ti,ab,kw
#5 (#1 or #2 or #3 or #4)
- #6 MeSH descriptor: [endoscopic mucosal resection] explode all trees
#7 (underwater or under water):ti,ab,kw
#8 (endoscopic mucosal resection or polypectomy or EMR or endoscopic resection or endoscopic mucosectom*):ti,ab,kw
#9 #7 AND (#6 OR #8)
#10

Appendix 2: MEDLINE (via PubMed) search strategy

- #1 colorect*[tiab]
#2 colon*[tiab]
#3 bowel[tiab]
#4 rect*[tiab]
#5 "Intestinal Mucosa"[tiab]
#6 polyp*[tiab]
#7 neoplas*[tiab]
#8 tumour*[tiab]
#9 tumor[tiab]
#10 adenom*[tiab]
#11 lesion*[tiab]
#12 carcinom*[tiab]
#13 adenocarcinom*[tiab]
#14 cancer*[tiab]
#15 #1 OR #2 OR #3 OR #4 OR #5
#16 #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #13 OR #14
#17 #15 AND #16
- #18 "polyps"[MeSH Terms]
#19 "neoplasms"[MeSH Terms]
#20 "intestines"[MeSH Terms]
#21 #18 OR #19 OR #20
#22 #17 OR #21
- #23 underwater[tiab]
#24 "under water"[tiab]
#25 endoscopic mucosal resection[MeSH]
#26 "endoscopic mucosal resection"[tiab]

#27 polypectomy[tiab]
 #28 EMR[tiab]
 #29 "endoscopic resection"[tiab]
 #30 "endoscopic mucossectom*"[tiab]
 #31 #23 OR #24
 #32 #25 OR #26 OR #27 OR #28 OR #29 OR #30
 #33 #31 AND #32

 #34 randomized controlled trial [pt]
 #35 controlled clinical trial [pt]
 #36 randomized [tiab]
 #37 placebo [tiab]
 #38 drug therapy [sh]
 #39 randomly [tiab]
 #40 trial [tiab]
 #41 groups [tiab]
 #42 #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41
 #43 animals [mh] NOT humans [mh]
 #44 #42 NOT #43

 #45 #22 AND #33 AND #44

Appendix 3: EMBASE (via ProQuest Dialog) search strategy

S1. (ab(colorect) OR ti(colorect)) AND (ab(polyp) OR ti(polyp))
 S2. (ab(colorect) OR ti(colorect)) AND (ab(polyps) OR ti(polyps))
 S3. (ab(colorectal) OR ti(colorectal)) AND (ab(polyp) OR ti(polyp))
 S4. (ab(colorectal) OR ti(colorectal)) AND (ab(polyps) OR ti(polyps))
 S5. (ab(colorect) OR ti(colorect)) AND (ab(neoplasm) OR ti(neoplasm))
 S6. (ab(colorect) OR ti(colorect)) AND (ab(neoplasms) OR ti(neoplasms))
 S7. (ab(colorectal) OR ti(colorectal)) AND (ab(neoplasm) OR ti(neoplasm))
 S8. (ab(colorectal) OR ti(colorectal)) AND (ab(neoplasms) OR ti(neoplasms))
 S9. (ab(colorect) OR ti(colorect)) AND (ab(tumour) OR ti(tumour))
 S10. (ab(colorect) OR ti(colorect)) AND (ab(tumours) OR ti(tumours))
 S11. (ab(colorectal) OR ti(colorectal)) AND (ab(tumour) OR ti(tumour))
 S12. (ab(colorectal) OR ti(colorectal)) AND (ab(tumours) OR ti(tumours))
 S13. (ab(colorect) OR ti(colorect)) AND (ab(tumor) OR ti(tumor))
 S14. (ab(colorect) OR ti(colorect)) AND (ab(tumors) OR ti(tumors))
 S15. (ab(colorectal) OR ti(colorectal)) AND (ab(tumor) OR ti(tumor))
 S16. (ab(colorectal) OR ti(colorectal)) AND (ab(tumors) OR ti(tumors))
 S17. (ab(colorectal) OR ti(colorectal)) AND (ab(adenom) OR ti(adenom))
 S18. (ab(colorectal) OR ti(colorectal)) AND (ab(adenoms) OR ti(adenoms))
 S19. (ab(colorectal) OR ti(colorectal)) AND (ab(lesion) OR ti(lesion))
 S20. (ab(colorectal) OR ti(colorectal)) AND (ab(lesions) OR ti(lesions))
 S21. (ab(colorect) OR ti(colorect)) AND (ab(carcinom) OR ti(carcinom))
 S22. (ab(colorectal) OR ti(colorectal)) AND (ab(carcinom) OR ti(carcinom))
 S23. (ab(colorectal) OR ti(colorectal)) AND (ab(carcinoms) OR ti(carcinoms))
 S24. (ab(colorectal) OR ti(colorectal)) AND (ab(adenocarcinom) OR ti(adenocarcinom))
 S25. (ab(colorectal) OR ti(colorectal)) AND (ab(adenocarcinoms) OR ti(adenocarcinoms))
 S26. (ab(colorect) OR ti(colorect)) AND (ab(cancer) OR ti(cancer))
 S27. (ab(colorect) OR ti(colorect)) AND (ab(cancers) OR ti(cancers))
 S28. (ab(colorectal) OR ti(colorectal)) AND (ab(cancer) OR ti(cancer))
 S29. (ab(colorectal) OR ti(colorectal)) AND (ab(cancers) OR ti(cancers))
 S30. (ab(colon) OR ti(colon)) AND (ab(polyp) OR ti(polyp))
 S31. (ab(colon) OR ti(colon)) AND (ab(polyps) OR ti(polyps))

S32. (ab(colonal) OR ti(colonal)) AND (ab(polyp) OR ti(polyp))
 S33. (ab(colonal) OR ti(colonal)) AND (ab(polyyps) OR ti(polyyps))
 S34. (ab(colon) OR ti(colon)) AND (ab(neoplasm) OR ti(neoplasm))
 S35. (ab(colon) OR ti(colon)) AND (ab(neoplasms) OR ti(neoplasms))
 S36. (ab(colonal) OR ti(colonal)) AND (ab(neoplasm) OR ti(neoplasm))
 S37. (ab(colonal) OR ti(colonal)) AND (ab(neoplasms) OR ti(neoplasms))
 S38. (ab(colon) OR ti(colon)) AND (ab(tumour) OR ti(tumour))
 S39. (ab(colon) OR ti(colon)) AND (ab(tumours) OR ti(tumours))
 S40. (ab(colonal) OR ti(colonal)) AND (ab(tumour) OR ti(tumour))
 S41. (ab(colonal) OR ti(colonal)) AND (ab(tumours) OR ti(tumours))
 S42. (ab(colon) OR ti(colon)) AND (ab(tumor) OR ti(tumor))
 S43. (ab(colon) OR ti(colon)) AND (ab(tumors) OR ti(tumors))
 S44. (ab(colonal) OR ti(colonal)) AND (ab(tumor) OR ti(tumor))
 S45. (ab(colonal) OR ti(colonal)) AND (ab(tumors) OR ti(tumors))
 S46. (ab(colon) OR ti(colon)) AND (ab(adenom) OR ti(adenom))
 S47. (ab(colon) OR ti(colon)) AND (ab(adenoms) OR ti(adenoms))
 S48. (ab(colon) OR ti(colon)) AND (ab(adenom) OR ti(adenom))
 S49. (ab(colon) OR ti(colon)) AND (ab(adenoms) OR ti(adenoms))
 S50. (ab(colon) OR ti(colon)) AND (ab(lesion) OR ti(lesion))
 S51. (ab(colon) OR ti(colon)) AND (ab(lesions) OR ti(lesions))
 S52. (ab(colonal) OR ti(colonal)) AND (ab(lesion) OR ti(lesion))
 S53. (ab(colonal) OR ti(colonal)) AND (ab(lesions) OR ti(lesions))
 S54. (ab(colonal) OR ti(colonal)) AND (ab(lesions) OR ti(lesions))
 S55. (ab(colon) OR ti(colon)) AND (ab(carcinom) OR ti(carcinom))
 S56. (ab(colon) OR ti(colon)) AND (ab(carcinom) OR ti(carcinom))
 S57. (ab(colon) OR ti(colon)) AND (ab(carcinom) OR ti(carcinom))
 S58. (ab(colon) OR ti(colon)) AND (ab(carcinoms) OR ti(carcinoms))
 S59. (ab(colon) OR ti(colon)) AND (ab(adenocarcinom) OR ti(adenocarcinom))
 S60. (ab(colon) OR ti(colon)) AND (ab(adenocarcinoms) OR ti(adenocarcinoms))
 S61. (ab(colon) OR ti(colon)) AND (ab(cancer) OR ti(cancer))
 S62. (ab(colon) OR ti(colon)) AND (ab(cancers) OR ti(cancers))
 S63. (ab(colonal) OR ti(colonal)) AND (ab(cancer) OR ti(cancer))
 S64. (ab(colonal) OR ti(colonal)) AND (ab(cancers) OR ti(cancers))
 S65. (ab(bowel) OR ti(bowel)) AND (ab(polyp) OR ti(polyp))
 S66. (ab(bowel) OR ti(bowel)) AND (ab(polyyps) OR ti(polyyps))
 S67. (ab(bowel) OR ti(bowel)) AND (ab(neoplasm) OR ti(neoplasm))
 S68. (ab(bowel) OR ti(bowel)) AND (ab(neoplasms) OR ti(neoplasms))
 S69. (ab(bowel) OR ti(bowel)) AND (ab(tumour) OR ti(tumour))
 S70. (ab(bowel) OR ti(bowel)) AND (ab(tumours) OR ti(tumours))
 S71. (ab(bowel) OR ti(bowel)) AND (ab(tumor) OR ti(tumor))
 S72. (ab(bowel) OR ti(bowel)) AND (ab(tumors) OR ti(tumors))
 S73. (ab(bowel) OR ti(bowel)) AND (ab(adenom) OR ti(adenom))
 S74. (ab(bowel) OR ti(bowel)) AND (ab(adenoms) OR ti(adenoms))
 S75. (ab(bowel) OR ti(bowel)) AND (ab(lesion) OR ti(lesion))
 S76. (ab(bowel) OR ti(bowel)) AND (ab(lesions) OR ti(lesions))
 S77. (ab(bowel) OR ti(bowel)) AND (ab(carcinom) OR ti(carcinom))
 S78. (ab(bowel) OR ti(bowel)) AND (ab(carcinoms) OR ti(carcinoms))
 S79. (ab(bowel) OR ti(bowel)) AND (ab(adenocarcinom) OR ti(adenocarcinom))
 S80. (ab(bowel) OR ti(bowel)) AND (ab(cancer) OR ti(cancer))
 S81. (ab(bowel) OR ti(bowel)) AND (ab(cancers) OR ti(cancers))
 S82. (ab(rect) OR ti(rect)) AND (ab(polyp) OR ti(polyp))
 S83. (ab(rect) OR ti(rect)) AND (ab(polyyps) OR ti(polyyps))
 S84. (ab(rectal) OR ti(rectal)) AND (ab(polyp) OR ti(polyp))
 S85. (ab(rectal) OR ti(rectal)) AND (ab(polyyps) OR ti(polyyps))

S86. (ab(rectal) OR ti(rectal)) AND (ab(polyp) OR ti(polyp))
S87. (ab(rectal) OR ti(rectal)) AND (ab(polyyps) OR ti(polyyps))
S88. (ab(rect) OR ti(rect)) AND (ab(neoplasm) OR ti(neoplasm))
S89. (ab(rect) OR ti(rect)) AND (ab(neoplasms) OR ti(neoplasms))
S90. (ab(rectum) OR ti(rectum)) AND (ab(neoplasm) OR ti(neoplasm))
S91. (ab(rectum) OR ti(rectum)) AND (ab(neoplasms) OR ti(neoplasms))
S92. (ab(rectum) OR ti(rectum)) AND (ab(polyp) OR ti(polyp))
S93. (ab(rectum) OR ti(rectum)) AND (ab(polyyps) OR ti(polyyps))
S94. (ab(rect) OR ti(rect)) AND (ab(tumour) OR ti(tumour))
S95. (ab(rect) OR ti(rect)) AND (ab(tumours) OR ti(tumours))
S96. (ab(rectal) OR ti(rectal)) AND (ab(tumour) OR ti(tumour))
S97. (ab(rectal) OR ti(rectal)) AND (ab(tumours) OR ti(tumours))
S98. (ab(rectum) OR ti(rectum)) AND (ab(tumour) OR ti(tumour))
S99. (ab(rectum) OR ti(rectum)) AND (ab(tumours) OR ti(tumours))
S100. (ab(rect) OR ti(rect)) AND (ab(tumor) OR ti(tumor))
S101. (ab(rect) OR ti(rect)) AND (ab(tumors) OR ti(tumors))
S102. (ab(rectal) OR ti(rectal)) AND (ab(tumor) OR ti(tumor))
S103. (ab(rectal) OR ti(rectal)) AND (ab(tumors) OR ti(tumors))
S104. (ab(rectum) OR ti(rectum)) AND (ab(tumor) OR ti(tumor))
S105. (ab(rectum) OR ti(rectum)) AND (ab(tumors) OR ti(tumors))
S106. (ab(rectal) OR ti(rectal)) AND (ab(adenom) OR ti(adenom))
S107. (ab(rectal) OR ti(rectal)) AND (ab(adenoms) OR ti(adenoms))
S108. (ab(rectum) OR ti(rectum)) AND (ab(adenom) OR ti(adenom))
S109. (ab(rectum) OR ti(rectum)) AND (ab(adenoms) OR ti(adenoms))
S110. (ab(rect) OR ti(rect)) AND (ab(lesion) OR ti(lesion))
S111. (ab(rect) OR ti(rect)) AND (ab(lesions) OR ti(lesions))
S112. (ab(rectal) OR ti(rectal)) AND (ab(lesion) OR ti(lesion))
S113. (ab(rectal) OR ti(rectal)) AND (ab(lesions) OR ti(lesions))
S114. (ab(rectum) OR ti(rectum)) AND (ab(lesion) OR ti(lesion))
S115. (ab(rectum) OR ti(rectum)) AND (ab(lesions) OR ti(lesions))
S116. (ab(rect) OR ti(rect)) AND (ab(carcinom) OR ti(carcinom))
S117. (ab(rectal) OR ti(rectal)) AND (ab(carcinom) OR ti(carcinom))
S118. (ab(rectal) OR ti(rectal)) AND (ab(carcinoms) OR ti(carcinoms))
S119. (ab(rectum) OR ti(rectum)) AND (ab(carcinom) OR ti(carcinom))
S120. (ab(rectum) OR ti(rectum)) AND (ab(carcinoms) OR ti(carcinoms))
S121. (ab(rect) OR ti(rect)) AND (ab(adenocarcinom) OR ti(adenocarcinom))
S122. (ab(rectal) OR ti(rectal)) AND (ab(adenocarcinom) OR ti(adenocarcinom))
S123. (ab(rectum) OR ti(rectum)) AND (ab(adenocarcinom) OR ti(adenocarcinom))
S124. (ab(rect) OR ti(rect)) AND (ab(cancer) OR ti(cancer))
S125. (ab(rect) OR ti(rect)) AND (ab(cancers) OR ti(cancers))
S126. (ab(rectal) OR ti(rectal)) AND (ab(cancer) OR ti(cancer))
S127. (ab(rectal) OR ti(rectal)) AND (ab(cancers) OR ti(cancers))
S128. (ab(rectum) OR ti(rectum)) AND (ab(cancer) OR ti(cancer))
S129. (ab(rectum) OR ti(rectum)) AND (ab(cancers) OR ti(cancers))
S130. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(polyp) OR ti(polyp))
S131. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(polyyps) OR ti(polyyps))
S132. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(neoplasm) OR ti(neoplasm))
S133. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(neoplasms) OR ti(neoplasms))
S134. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(tumour) OR ti(tumour))
S135. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(tumours) OR ti(tumours))
S136. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(tumor) OR ti(tumor))
S137. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(tumors) OR ti(tumors))
S138. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(adenom) OR ti(adenom))
S139. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(lesion) OR ti(lesion))

S140. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(lesions) OR ti(lesions))
 S141. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(carcinom) OR ti(carcinom))
 S142. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(cancer) OR ti(cancer))
 S143. (ab(intestinal mucosa) OR ti(intestinal mucosa)) AND (ab(cancers) OR ti(cancers))
 S144. (ab(colorectum) OR ti(colorectum)) AND (ab(polyp) OR ti(polyp))
 S145. (ab(colorectum) OR ti(colorectum)) AND (ab(polyps) OR ti(polyps))
 S146. (ab(colorectum) OR ti(colorectum)) AND (ab(neoplasm) OR ti(neoplasm))
 S147. (ab(colorectum) OR ti(colorectum)) AND (ab(neoplasms) OR ti(neoplasms))
 S148. (ab(colorectum) OR ti(colorectum)) AND (ab(tumour) OR ti(tumour))
 S149. (ab(colorectum) OR ti(colorectum)) AND (ab(tumours) OR ti(tumours))
 S150. (ab(colorectum) OR ti(colorectum)) AND (ab(tumor) OR ti(tumor))
 S151. (ab(colorectum) OR ti(colorectum)) AND (ab(tumors) OR ti(tumors))
 S152. (ab(colorectum) OR ti(colorectum)) AND (ab(lesion) OR ti(lesion))
 S153. (ab(colorectum) OR ti(colorectum)) AND (ab(lesions) OR ti(lesions))
 S154. (ab(colorectum) OR ti(colorectum)) AND (ab(cancer) OR ti(cancer))
 S155. (ab(colorectum) OR ti(colorectum)) AND (ab(cancers) OR ti(cancers))
 S156. EMB.EXACT.EXPLODE("polyp")
 S157. EMB.EXACT.EXPLODE("neoplasm")
 S158. EMB.EXACT.EXPLODE("intestine")
 S159. S156 OR S157 OR S158
 S160. S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR
 S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24
 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR
 S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43 OR S44 OR S45 OR S46 OR S47
 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54 OR S55 OR S56 OR S57 OR S58 OR
 S59 OR S60 OR S61 OR S62 OR S63 OR S64 OR S65 OR S66 OR S67 OR S68 OR S69 OR S70
 OR S71 OR S72 OR S73 OR S74 OR S75 OR S76 OR S77 OR S78 OR S79 OR S80 OR S81 OR
 S82 OR S83 OR S84 OR S85 OR S86 OR S87 OR S88 OR S89 OR S90 OR S91 OR S92 OR S93
 OR S94 OR S95 OR S96 OR S97 OR S98 OR S99 OR S100 OR S101 OR S102 OR S103 OR S104
 OR S105 OR S106 OR S107 OR S108 OR S109 OR S110 OR S111 OR S112 OR S113 OR S114
 OR S115 OR S116 OR S117 OR S118 OR S119 OR S120 OR S121 OR S122 OR S123 OR S124
 OR S125 OR S126 OR S127 OR S128 OR S129 OR S130 OR S131 OR S132 OR S133 OR S134
 OR S135 OR S136 OR S137 OR S138 OR S139 OR S140 OR S141 OR S142 OR S143 OR S144
 OR S145 OR S146 OR S147 OR S148 OR S149 OR S150 OR S151 OR S152 OR S153 OR S154
 OR S155
 S161. S159 OR S160
 S162. ab(underwater) OR ti(underwater)

Appendix 4: ICTRP search strategy

Condition: ("colorectal" OR "colorectum" OR "colon" OR "bowel" OR "rectal" OR "rectum") AND
 ("polyp" OR "neoplasia" OR "adenoma" OR "lesion" OR "carcinoma" OR "adenocarcinoma")
 AND

Intervention: "UEMR" OR ("underwater" AND ("endoscopic mucosal resection" OR "polypectomy"
 OR "EMR" OR "endoscopic resection" OR "endoscopic mucosectomy"))

Appendix 5: ClinicalTrials.gov search strategy

Condition or disease: (colorectal OR colorectum OR colon OR bowel OR rectal OR rectum) AND
 (polyp OR neoplasia OR adenoma OR lesion OR carcinoma OR adenocarcinoma)

Intervention: UEMR OR (underwater AND ((endoscopic mucosal resection) OR polypectomy OR
 EMR OR (endoscopic resection) OR (endoscopic mucosectomy)))

